

Roberto Devesa Fernández

Platform Engineer

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Up-to-date version of CV is available at
<https://Roberdvs.github.io/cv-repository>

Residence	 Asturias, Spain
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Platform Engineer with a background in System Administration.

Over the past years, I have been busy designing, implementing and evolving a Kubernetes-based Internal Developer Platform, acting as technical lead for a small team of engineers and championing DevSecOps & SRE practices; fostering a culture of continuous improvement and full-service ownership.

I'm used to creating reliable and scalable systems, streamlining processes, dealing with migrations from legacy systems to modern and simpler alternatives, optimizing infrastructure, and addressing cross-cutting concerns, primarily leveraging cloud services and open-source tooling while following Infrastructure as Code and GitOps approaches.

Familiar with being part of an on-call rotation, supporting mission-critical services and infrastructure, and following incident response best practices like writing blameless postmortems.

Professional Experience

Mar 2022 – Present. Remote

Staff Platform Engineer

Continue leading the evolution and roadmap of the Internal Developer Platform, nurturing a team around it as a tech lead while mentoring colleagues and teams on DevOps, SRE, and full-service ownership practices while working on organization-wide challenges.

I'm currently working on optimizing cloud costs by moving the vast majority of workloads to ARM architecture.

- Optimized infrastructure utilization by migrating from Kubernetes Cluster Autoscaler to Karpenter, reducing costs by 40%.
- Enhanced API performance and reduced infrastructure costs by collaborating with application teams to implement caching policies in our CDN.
- Spearheaded the migration from PagerDuty to BetterUptime for incident response.
- Automated Terraform code quality and security checks using GitHub Actions.
- Deployed an internal JupyterHub platform to provide Jupyter Notebooks for Data Engineers and Data Scientists.
- Collaborated on designing a versatile solution to serve multiple Large Language Models using LiteLLM, switching seamlessly between them without requiring modifications to applications.
- Helped the team migrate the majority of workloads to ARM architecture for increased cost savings.

DevOps Terraform Kubernetes AWS EKS

Feb 2020 - Mar 2022. Hybrid

Senior Platform Engineer

As the company grew to +100 engineers, established the formation of the Platform Engineering Team as the next step in our DevOps journey; a team responsible for promoting DevOps practices at scale and addressing cross-cutting concerns by providing application teams with an Internal Developer Platform that they can leverage to develop, deploy and operate their apps in a self-service fashion.

- Led the creation and evolution of the Kubernetes-based Internal Developer Platform, leveraging AWS EKS, IaC and open source tooling like Rancher, ArgoCD, Harbor, Prometheus, Grafana, ElasticSearch... while integrating Single Sign-on + RBAC.
- Migrated workloads from AWS ECS to AWS EKS, using Helm charts and GitOps practices.
- Architected and implemented a cloud-agnostic solution for running Big Data Spark Jobs on our Kubernetes platform, leveraging the spark-operator, Argo Workflows, and autoscaling capabilities, migrating from a previous AWS EMR setup and reducing vendor lock-in.
- Simplified the company's AWS user management by leveraging AWS SSO and integrating Google Workspace as Identity Provider.
- Improved security posture by integrating AWS WAF.
- Drove infrastructure cost savings through increased use of spot instances.
- Developed opinionated Terraform modules with embedded best practices.
- Led the migration of legacy systems to modern, simpler & more maintainable alternatives, like moving from an in-house VPN solution to Tailscale.

DevOps Terraform Kubernetes AWS EKS

Nov 2016 - Feb 2020

DevOps Engineer

Being a small company of ~20 people I used to wear many hats without a clearly defined role and help where I could, ranging from office IT to managing cloud infrastructure and everything in between, but I always gravitated more to DevOps-related tasks from which I'd highlight:

- Migrated from Atlassian Bamboo to Jenkins for Continuous Integration; configuring pipelines as code with reusable libraries among projects.
- Integrated new tools like SonarQube into the CI/CD process to improve SDLC.
- Managed cloud resources with Terraform and implemented collaborative IaC workflows using Atlantis.
- Worked on migrating static websites to simpler, cheaper, and more reliable hosting using S3 + CloudFront.
- Containerized applications with Docker and orchestrated deployments with AWS ECS.
- Improved scalability and reliability of backend services by applying SRE practices and leveraging Prometheus, Grafana, and the Elastic Stack.
- Implemented auto-scaling on our backend services to automatically adjust the capacity based on demand.
- Played a key role in scaling infrastructure to support high traffic during major events like Black Friday, while reducing the manual steps involved in the process of dealing with subsequent large-scale events.
- Be part of an on-call rotation, resolving incidents and writing blameless post-mortems, figuring out root causes, and taking action to prevent them from happening again.

DevOps Terraform AWS

Education

Bachelor's Degree in Telecommunications Engineering at Universidad de Oviedo [2008 - 2013]

Additional Experience